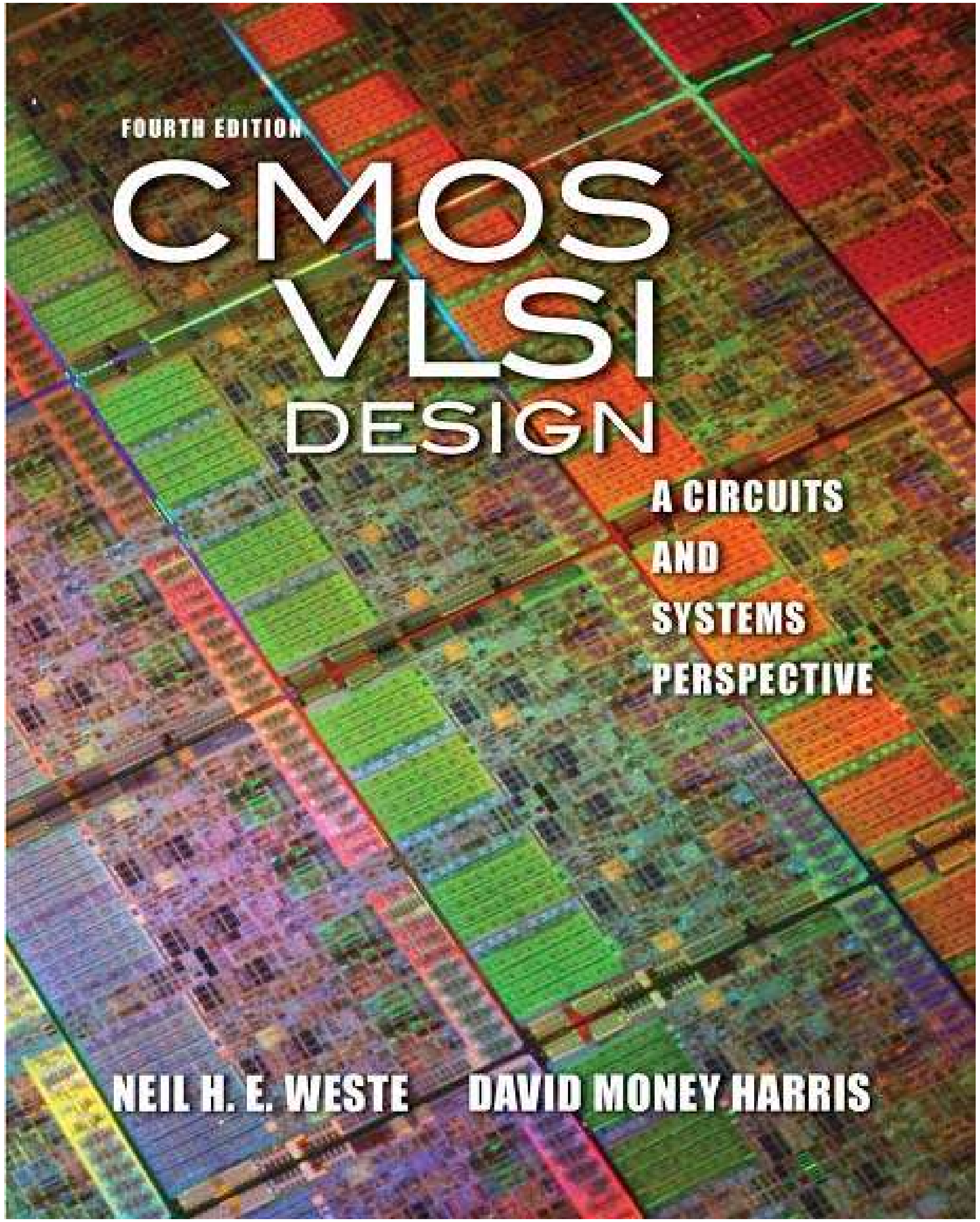
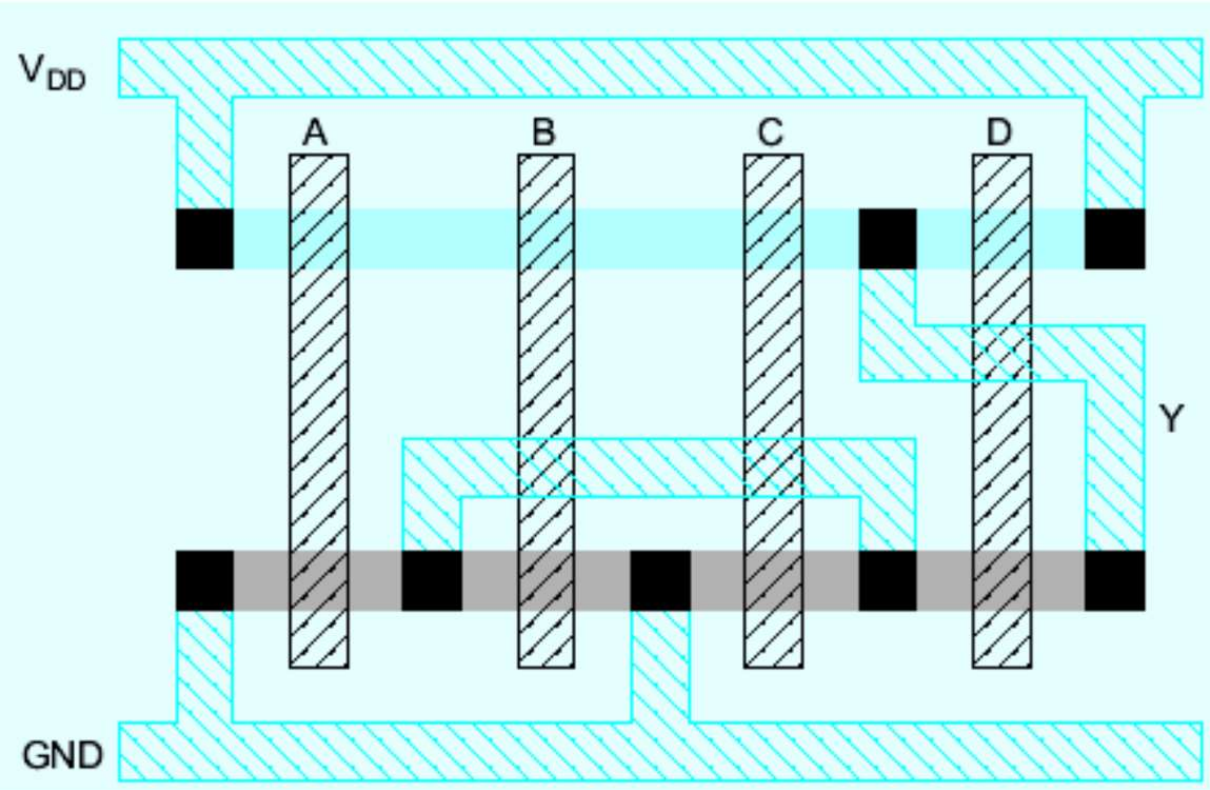
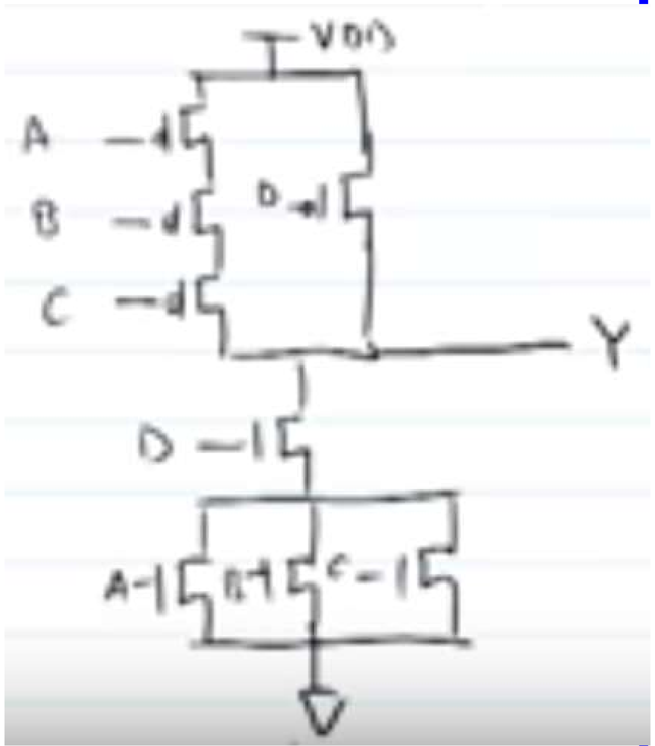


Chapter 5: Delay

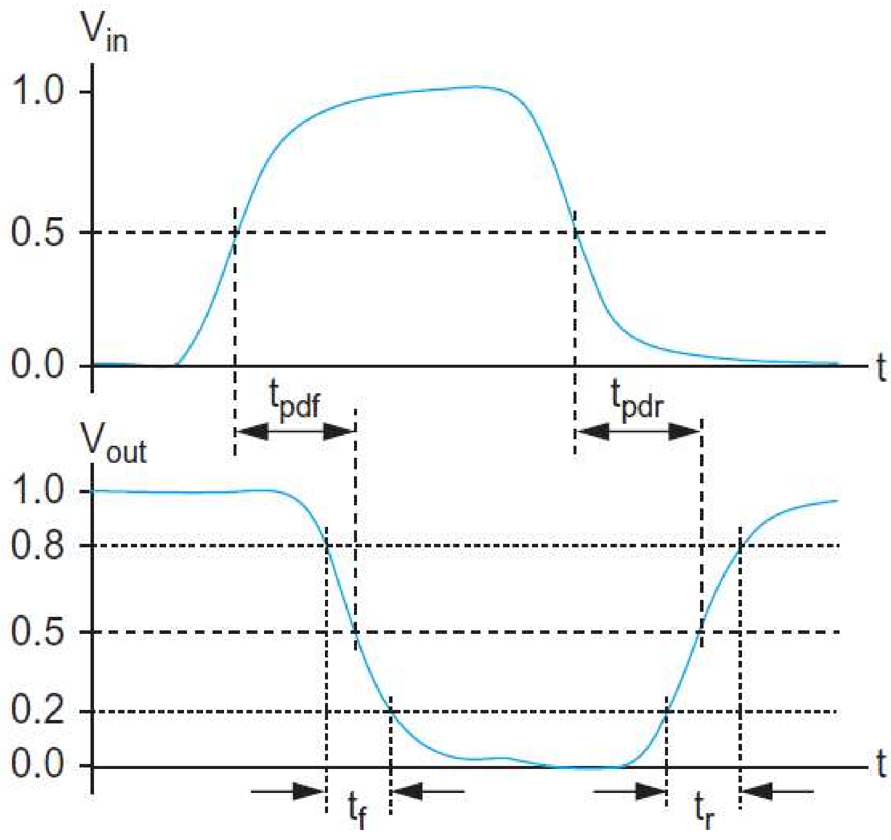
1. Delay definition
2. RC delay models
3. Linear delay models


V$_{DD}$
A
B
C
D
Y
GND


A
B
C
D
Y
D

1. Delay Definitions

- t_{pdr}: *rising propagation delay*
  - From input to rising output crossing $V_{DD}/2$
- t_{pdf}: *falling propagation delay*
  - From input to falling output crossing $V_{DD}/2$
- t_{pd}: *average propagation delay*
  - $t_{pd} = (t_{pdr} + t_{pdf})/2$
- t_r: *rise time*
  - From output crossing 0.2 V_{DD} to 0.8 V_{DD}
- t_f: *fall time*
  - From output crossing 0.8 V_{DD} to 0.2 V_{DD}



1. Delay Definitions

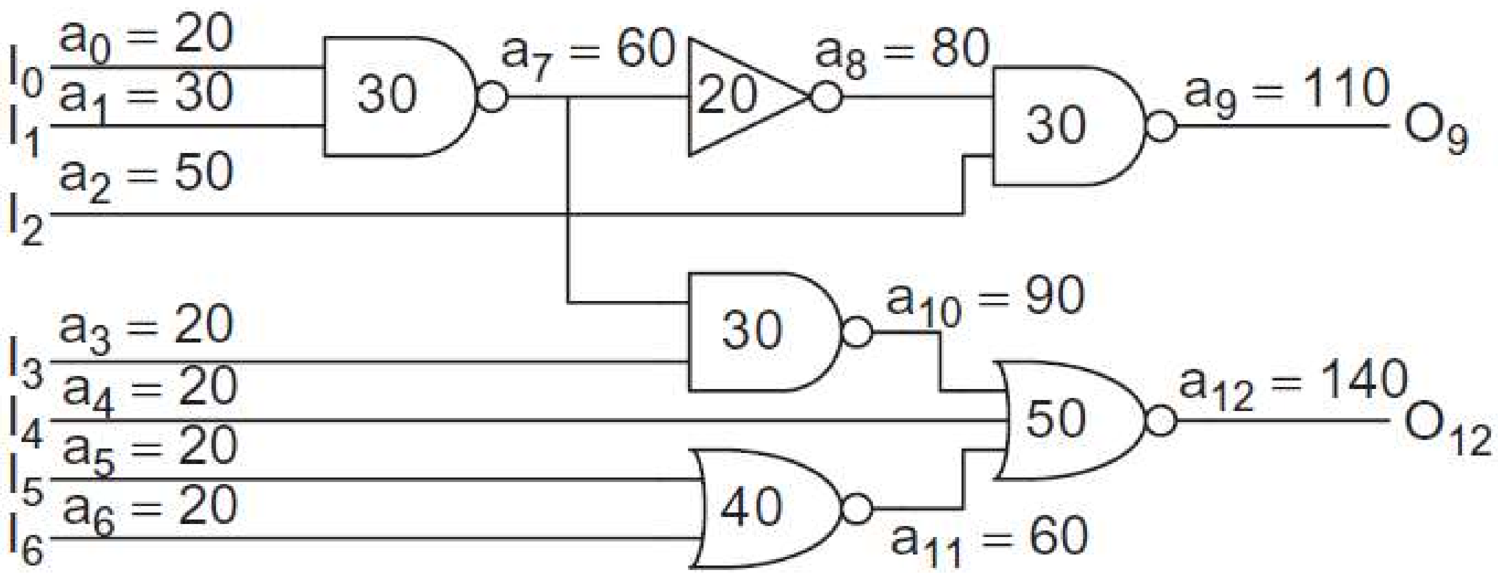
- **t_{cdr}**: *rising contamination delay*
  – From input to rising output crossing $V_{DD}/2$
- **t_{cdf}**: *falling contamination delay*
  – From input to falling output crossing $V_{DD}/2$
- **t_{cd}**: *average contamination delay*
  – $t_{cd} = (t_{cdr} + t_{cdf})/2$

1. Delay Definitions

The contamination delay, t_{cd}, is the fastest that the logic gate will change output based on a changed input.
The propagation delay, t_{pd}, is the slowest that the logic gate will change output based on a changed input.

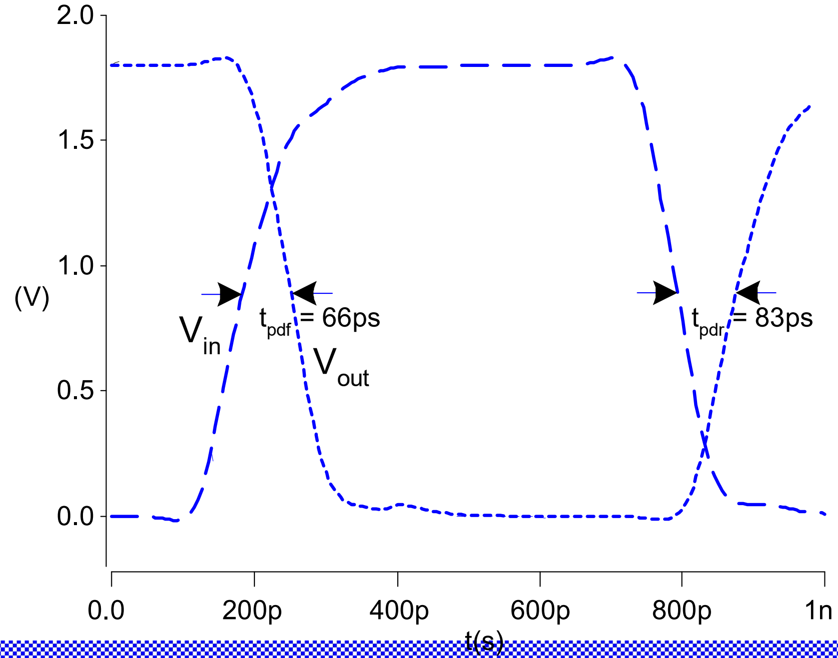
Arrival time

- Arrival time is the latest time at which each node in a block of logic will switch
- The *slack* is the difference between the required and arrival times.
- *Positive slack* means that the circuit meets timing.
- *Negative slack* means that the circuit is not fast enough.



Simulated Inverter Delay

- Solving differential equations by hand is too hard
- SPICE simulator solves the equations numerically
  - Uses more accurate I-V models too!
- But simulations take time to write, may hide insight



Delay Estimation

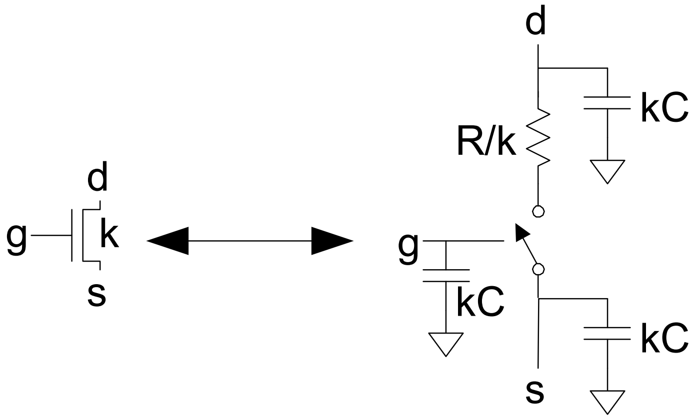
- We would like to be able to easily estimate delay
  - Not as accurate as simulation
  - But easier to ask "What if?"
- The step response usually looks like a 1st order RC response with a decaying exponential.
- Use RC delay models to estimate delay
  - C = total capacitance on output node
  - Use *effective resistance* R
  - So that $t_{pd} = RC$
- Characterize transistors by finding their effective R
  - Depends on average current as gate switches

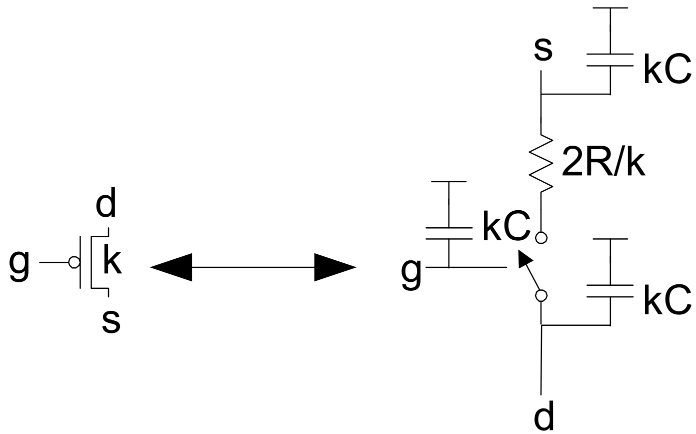
Effective Resistance

- Shockley models have limited value
  - Not accurate enough for modern transistors
  - Too complicated for much hand analysis
- Simplification: treat transistor as resistor
  - Replace $I_{ds}(V_{ds}, V_{gs})$ with effective resistance R
    - $I_{ds} = V_{ds}/R$
  - R averaged across switching of digital gate
- Too inaccurate to predict current at any given time
  - But good enough to predict RC delay

3. RC Delay Model

- ❑ Use equivalent circuits for MOS transistors
  - – Ideal switch + capacitance and ON resistance
  - – Unit nMOS has resistance R, capacitance C
  - – Unit pMOS has resistance 2R, capacitance C
- ❑ Capacitance proportional to width
- ❑ Resistance inversely proportional to width



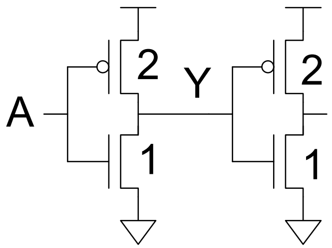


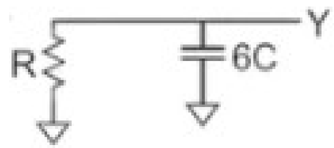
RC Values

- Capacitance
  - $C = C_g = C_s = C_d = 2$ fF/μm of gate width in 0.6 μm
  - Gradually decline to 1 fF/μm in nanometer techs.
- Resistance
  - $R \approx 6$ KΩ*μm in 0.6 μm process
  - Improves with shorter channel lengths
- Unit transistors
  - May refer to minimum contacted device (4/2 λ)
  - Or maybe 1 μm wide device
  - Doesn't matter as long as you are consistent

Inverter Delay Estimate

- Estimate the delay of a fanout-of-1 inverter

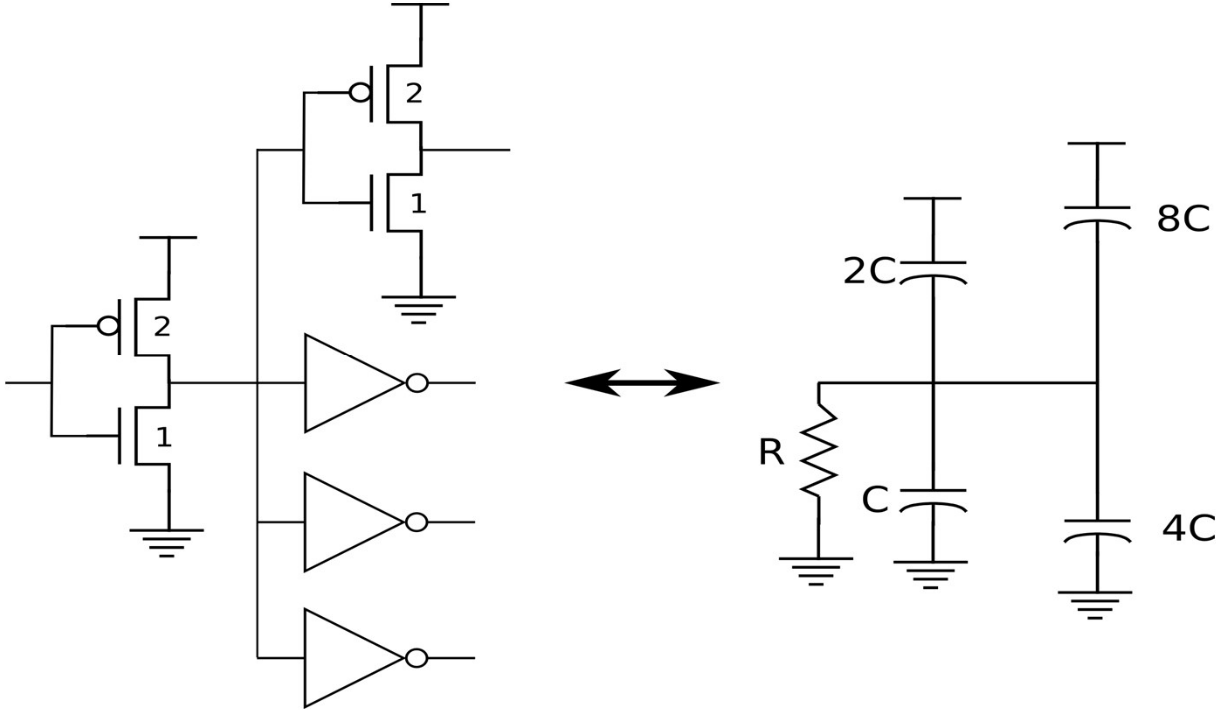




d = 6RC

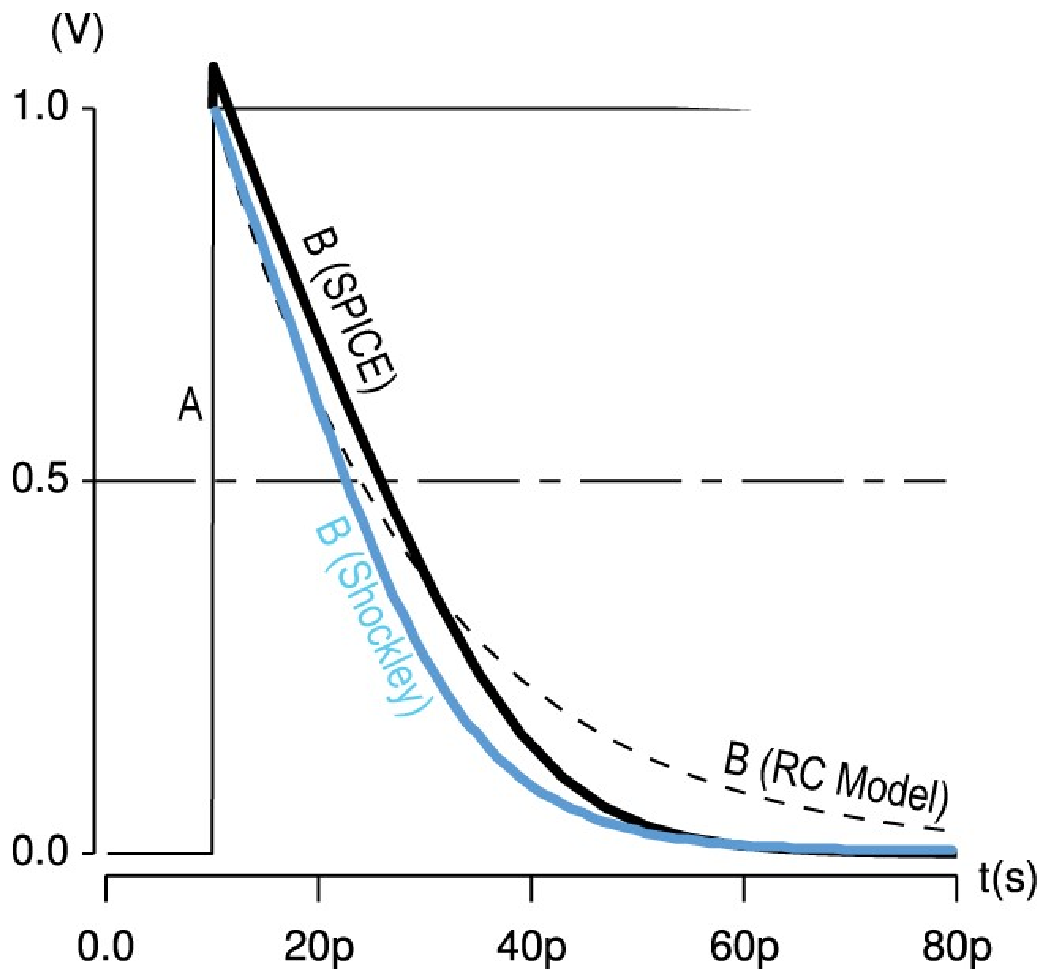
Inverter Delay Estimate

- Estimate the delay of an inverter driving 4 identical inverters – Fanout-of-4 (FO4) delay
- An important abstraction at higher levels of the design



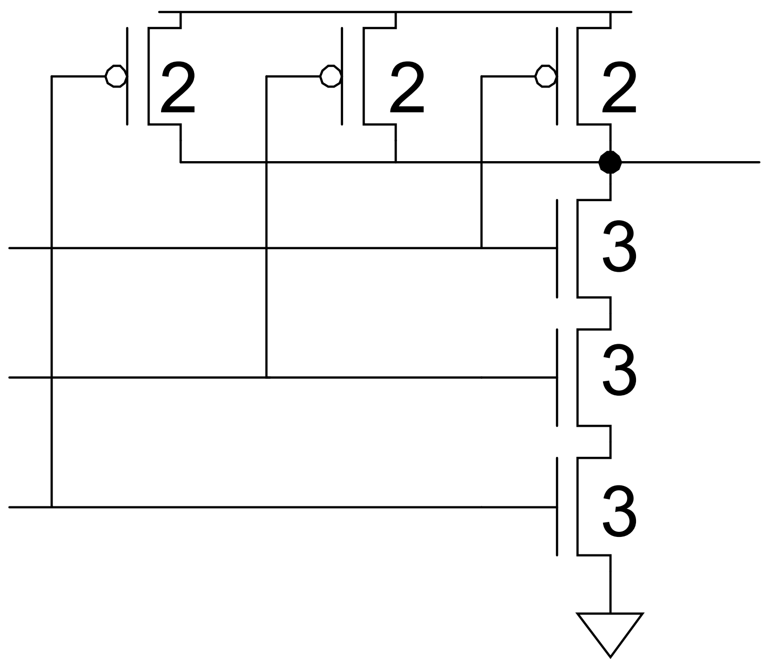
d = 15RC

Delay Model Comparison



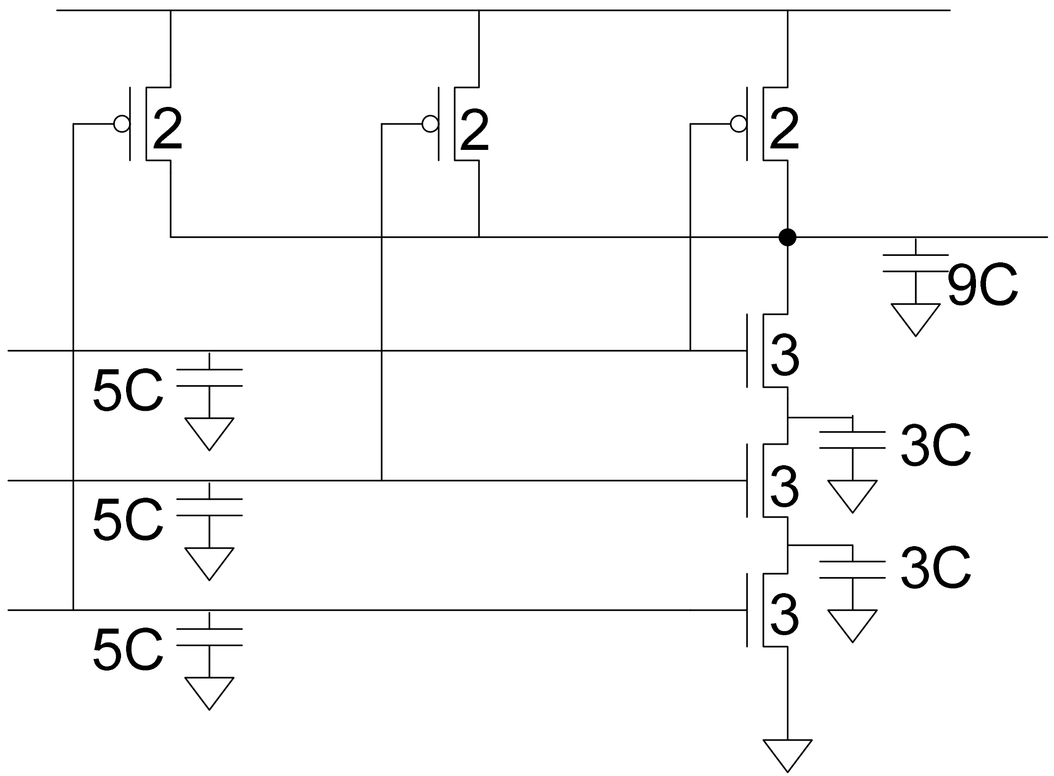
Example: 3-input NAND

- Sketch a 3-input NAND with transistor widths chosen to achieve effective rise and fall resistances equal to a unit inverter (R).



3-input NAND Caps

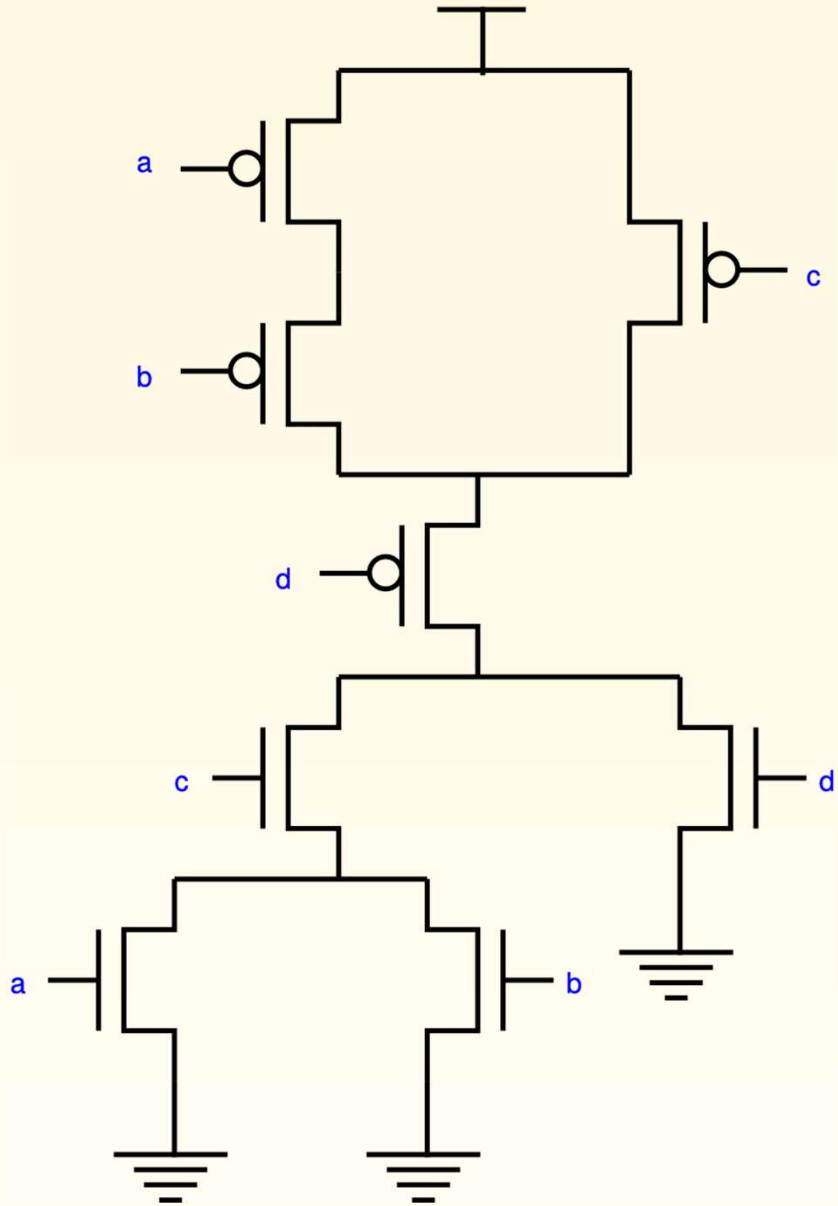
- Annotate the 3-input NAND gate with gate and diffusion capacitance.



Example: Sizing Complex Gate

Size the transistors in the circuit below so that it has the same drive strength, in the worst case, as an inverter that has PW = 5 and NW = 3.
Use the smallest widths possible to achieve this ratio.

Note: if there are multiple paths through a transistor, use the size for the "worst-case" input combination.

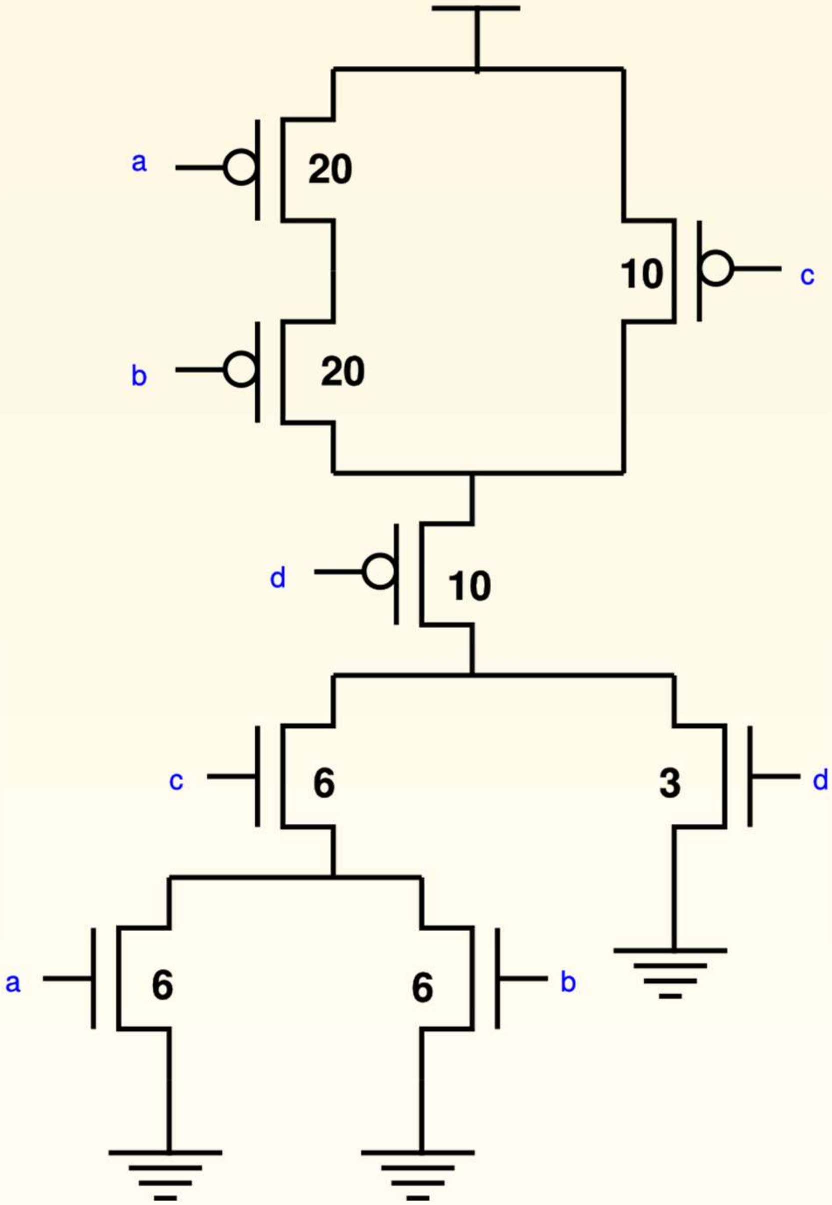


Example: Sizing Complex Gate

Size the transistors in the circuit below so that it has the same drive strength, in the worst case, as an inverter that has PW = 5 and NW = 3.
Use the smallest widths possible to achieve this ratio.

This solution does NOT use the smallest widths

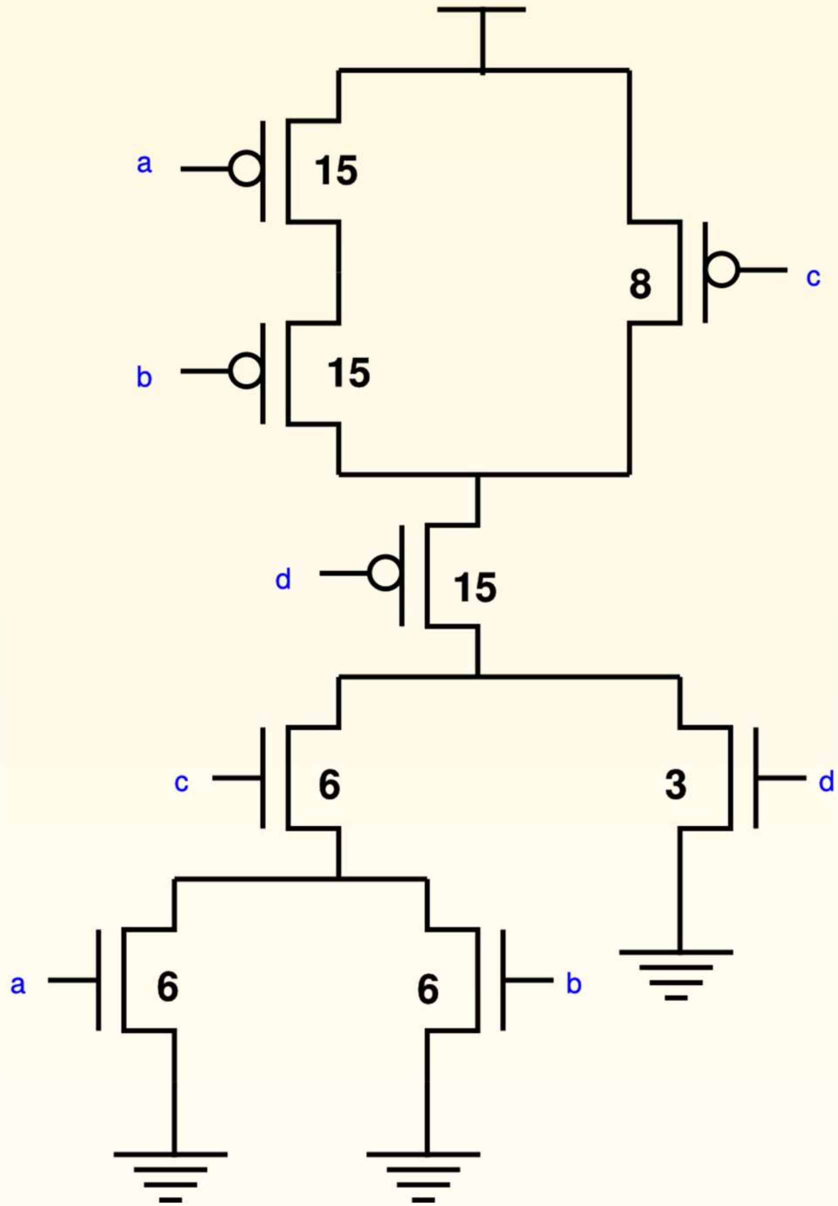
Note: if there are multiple paths through a transistor, use the size for the "worst-case" input combination.



Example: Sizing of Complex Gate – Better Solution

Size the transistors in the circuit below so that it has the same drive strength, in the worst case, as an inverter that has PW = 5 and NW = 3.
Use the smallest widths possible to achieve this ratio.

Note: if there are multiple paths through a transistor, use the size for the "worst-case" input combination.

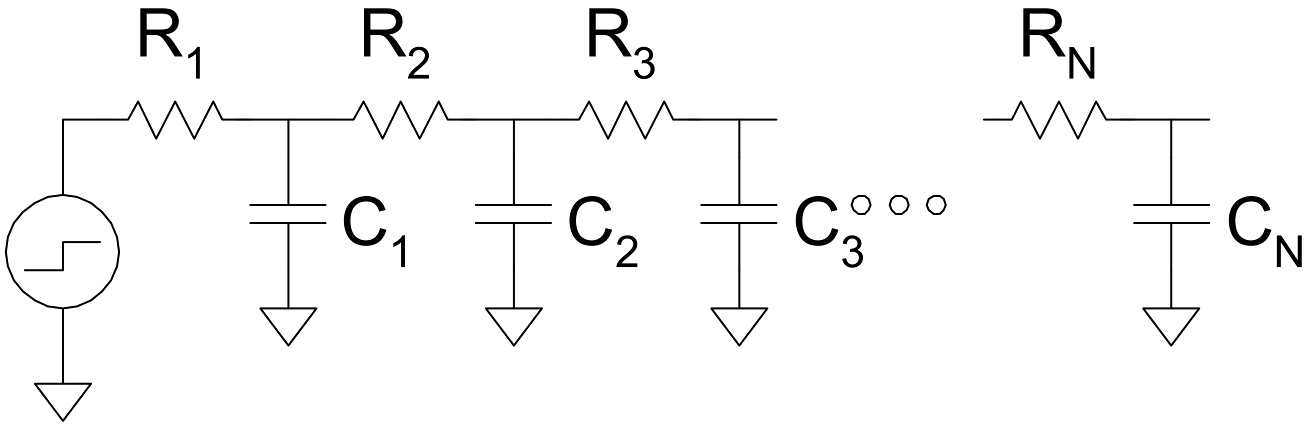


Elmore Delay

- ON transistors look like resistors
- Pullup or pulldown network modeled as *RC ladder*
- Elmore delay of RC ladder

$$t_{pd} \approx \sum_{\text{nodes } i} R_{i-to-source} C_i$$

$$= R_1 C_1 + \left(R_1 + R_2\right) C_2 + \ldots + \left(R_1 + R_2 + \ldots + R_N\right) C_N$$

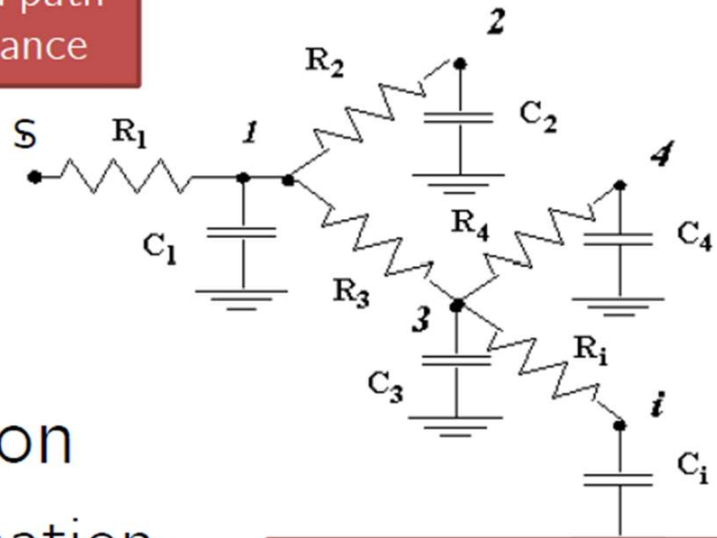


Elmore Delay

- Elmore delay at node i is given by

$$\tau_{Di} = \sum_{k=1}^{N} C_k R_{ik}$$

shared path resistance

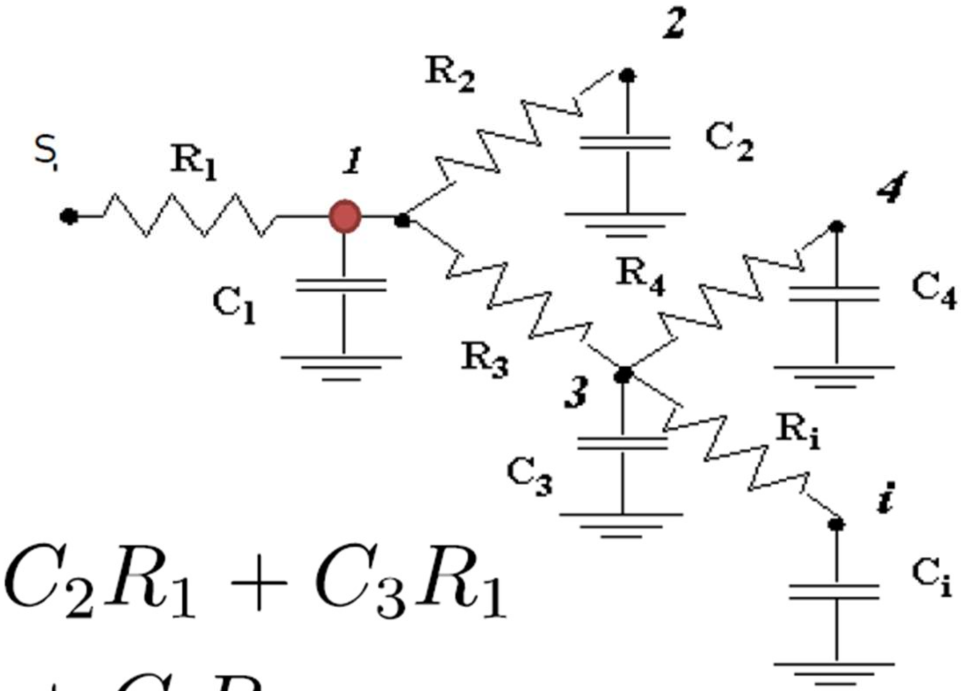


- First-order approximation
  - so-called Padé approximation

best approximation of a function by a rational function of given order

Elmore Delay – Ex.1

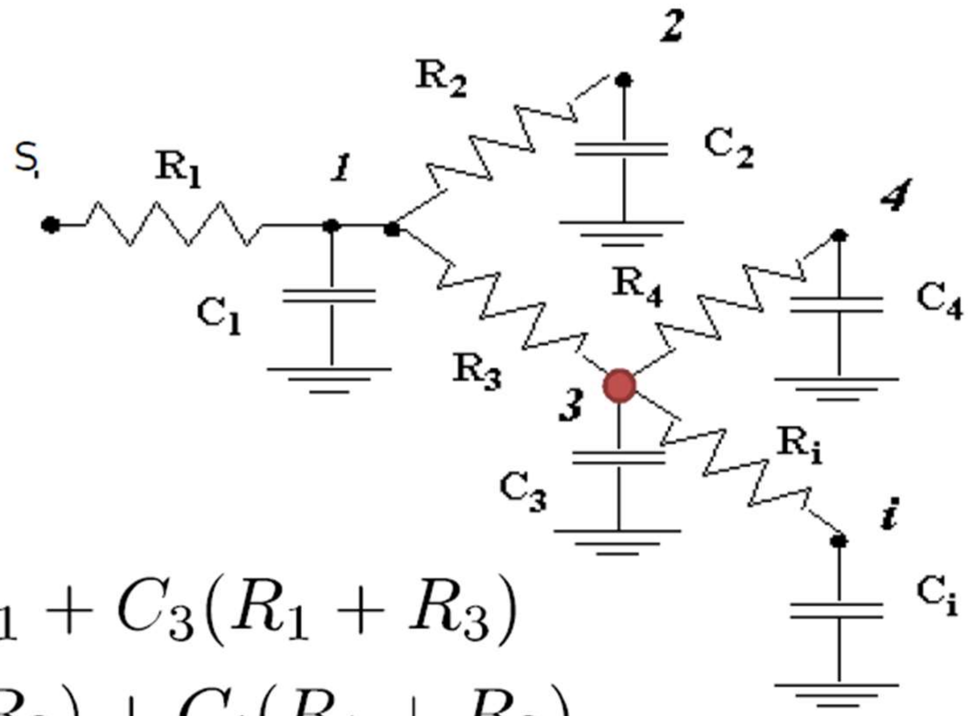
- Elmore delay τ_{D1} for node 1?



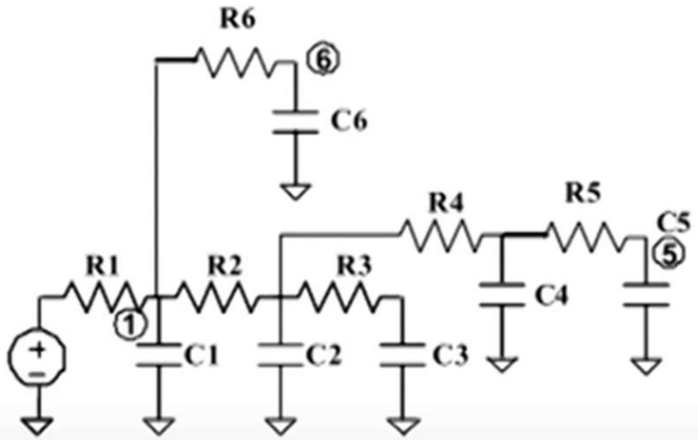
$$\tau_{D1} = C_1 R_1 + C_2 R_1 + C_3 R_1 + C_4 R_1 + C_i R_1$$

Elmore Delay – Ex.1

- Elmore delay τ_{D3} for node 3?



$$\tau_{D3} = C_1R_1 + C_2R_1 + C_3(R_1 + R_3) + C_4(R_1 + R_3) + C_i(R_1 + R_3)$$



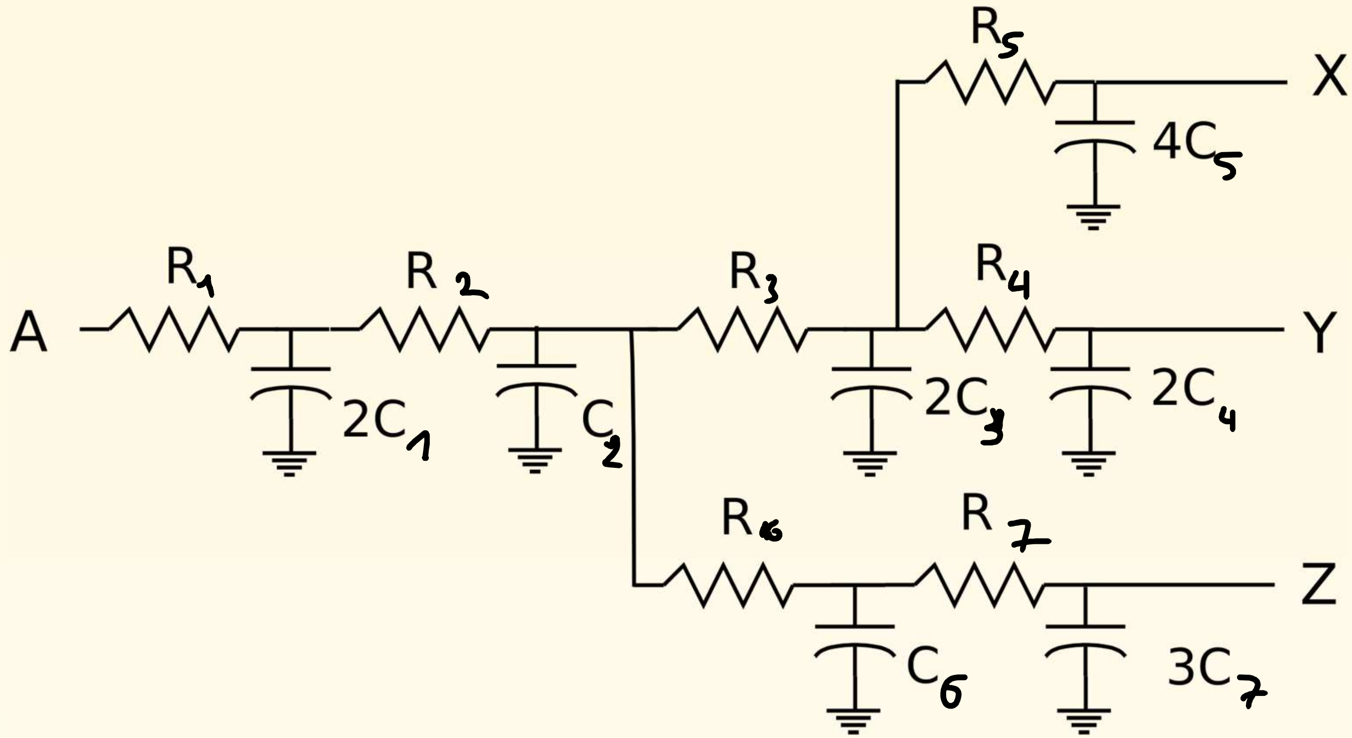
Compute the elmore delay at:

A) Node 1

B) Node 5

C) Node 6

Example: Elmore Delay Calculation

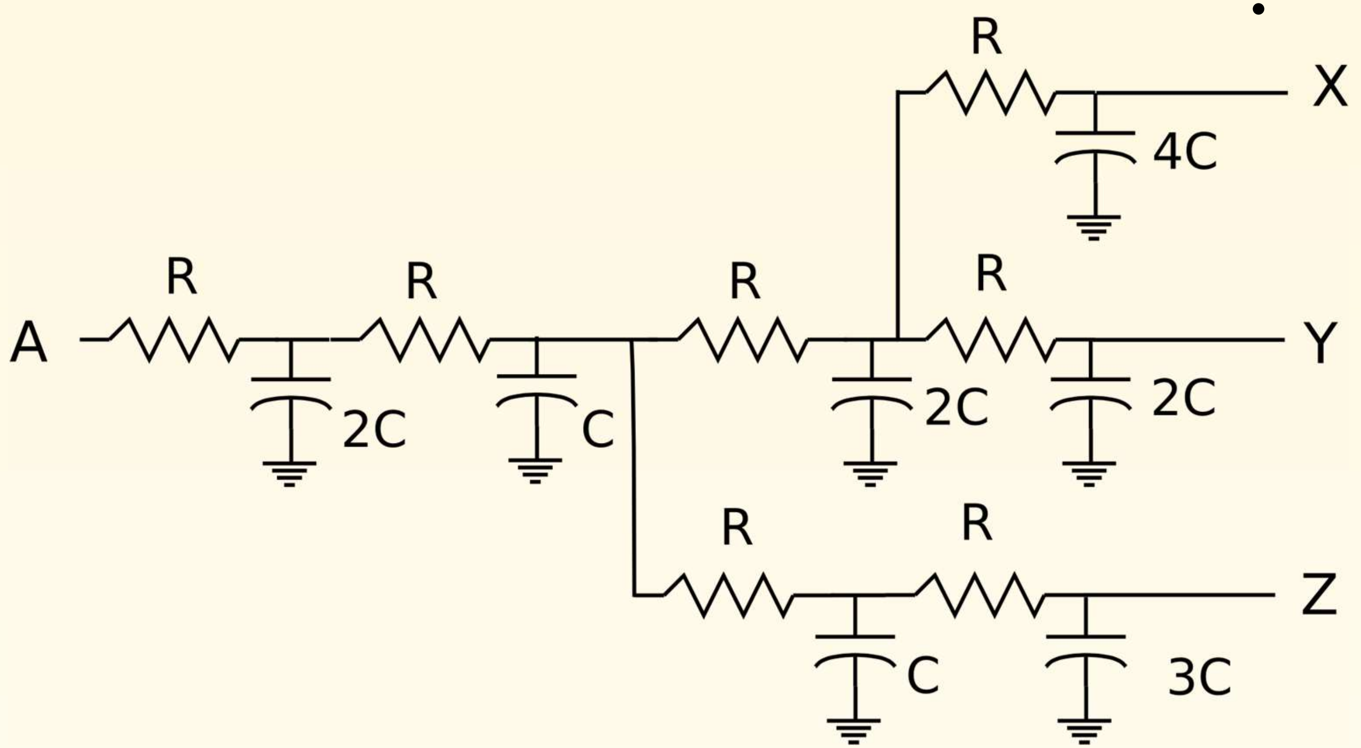


Delay from A to X: $C_1R_1 + C_2(R_{12}) + C_3(R_{123}) + C_4(R_{123}) + C_5(R_{1235}) + C_6(R_{12}) + C_7(R_{12}) = 40RC$

Delay from A to Y: $C_1R_1 + C_2(R_{12}) + C_3(R_{123}) + C_4(R_{1234}) + C_5(R_{123}) + C_6(R_{12}) + C_7(R_{12}) = 38RC$

Delay from A to Z: $C_1R_1 + C_2(R_{12}) + C_3(R_{12}) + C_4(R_{12}) + C_5(R_{12}) + C_6(R_{126}) + C_7(R_{1267}) = 35RC$

Example: Elmore Delay Calculation, Cont'd



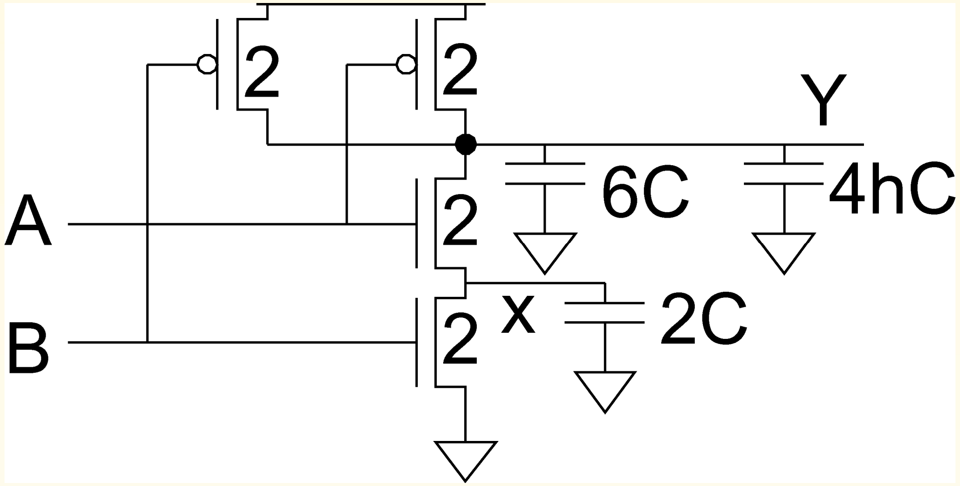
Delay from A to X: 40RC

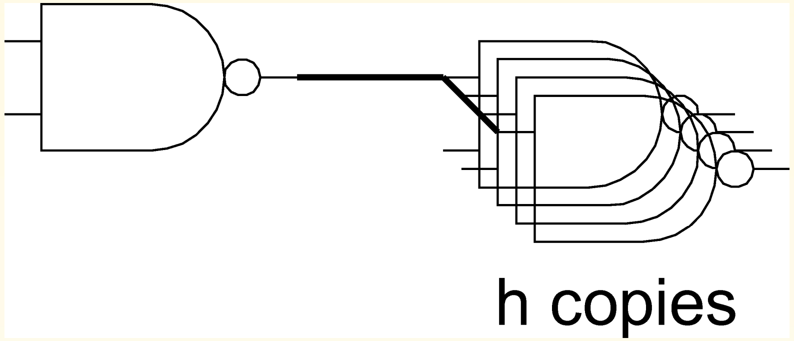
Delay from A to Y: 38RC

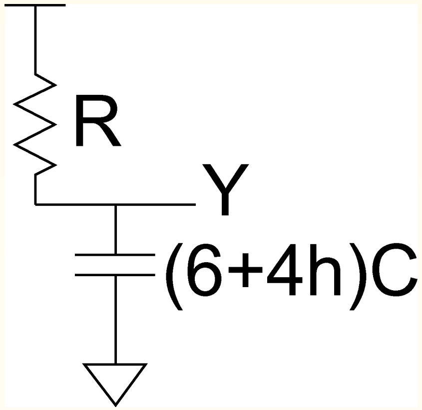
Delay from A to Z: 35RC

Example: Delay of 2-Input NAND Using Elmore Formulation

Estimate **rising** and falling propagation delays of a 2-input NAND driving h identical gates



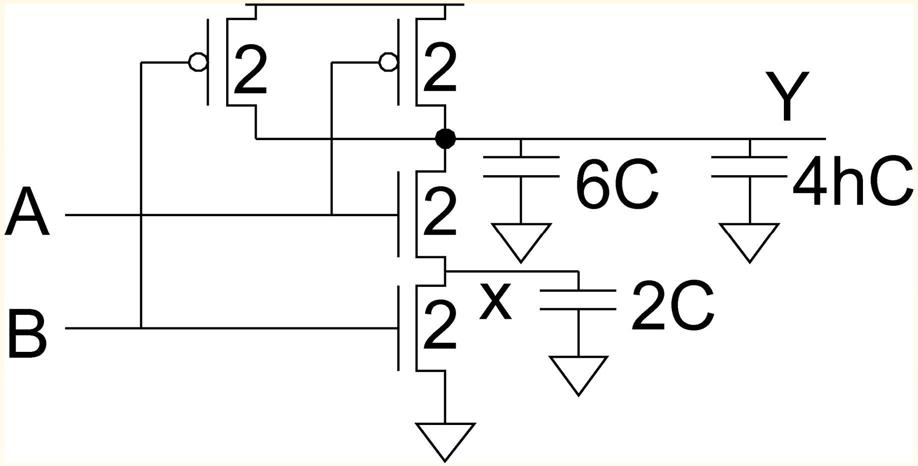


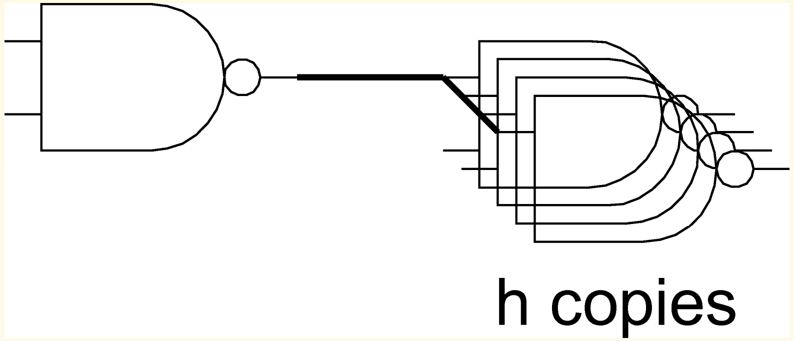


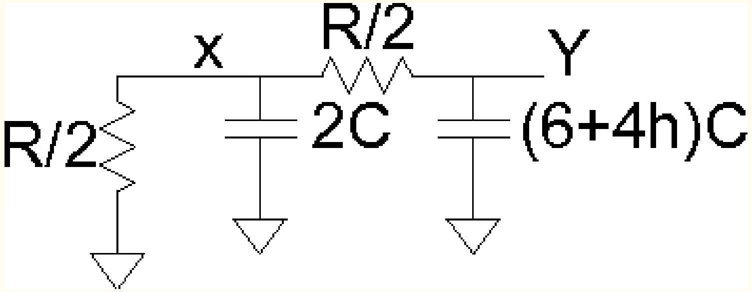
$$t_{pdr} = (6 + 4h)RC$$

Example: Delay of 2-Input NAND Using Elmore Formulation

Estimate rising and **falling** propagation delays of a 2-input NAND driving h identical gates



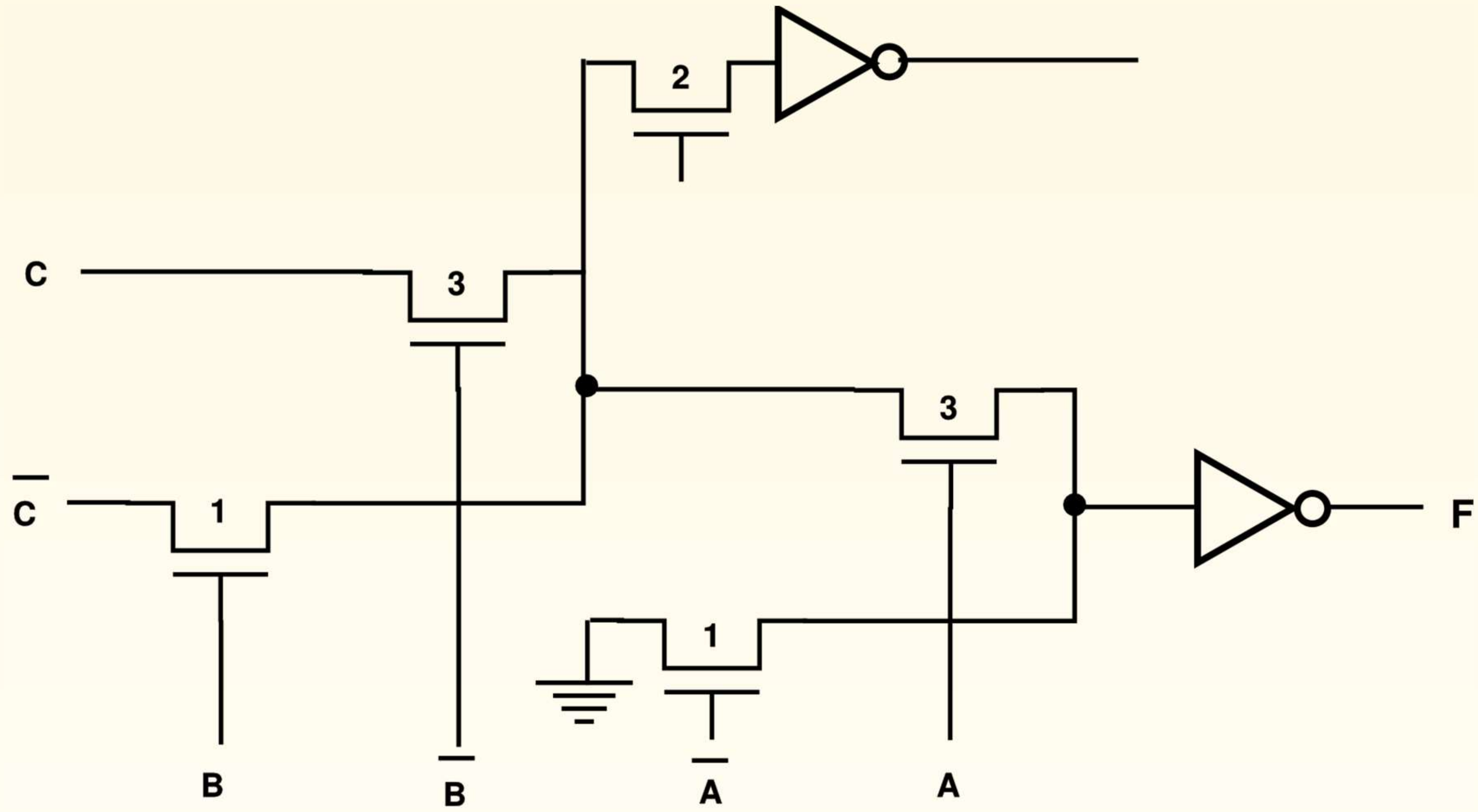




$$t_{pdf} = (2C)\frac{R}{2} + \left[(6+4h)C\right]\left(\frac{R}{2}+\frac{R}{2}\right)$$
$$= (7+4h)RC$$

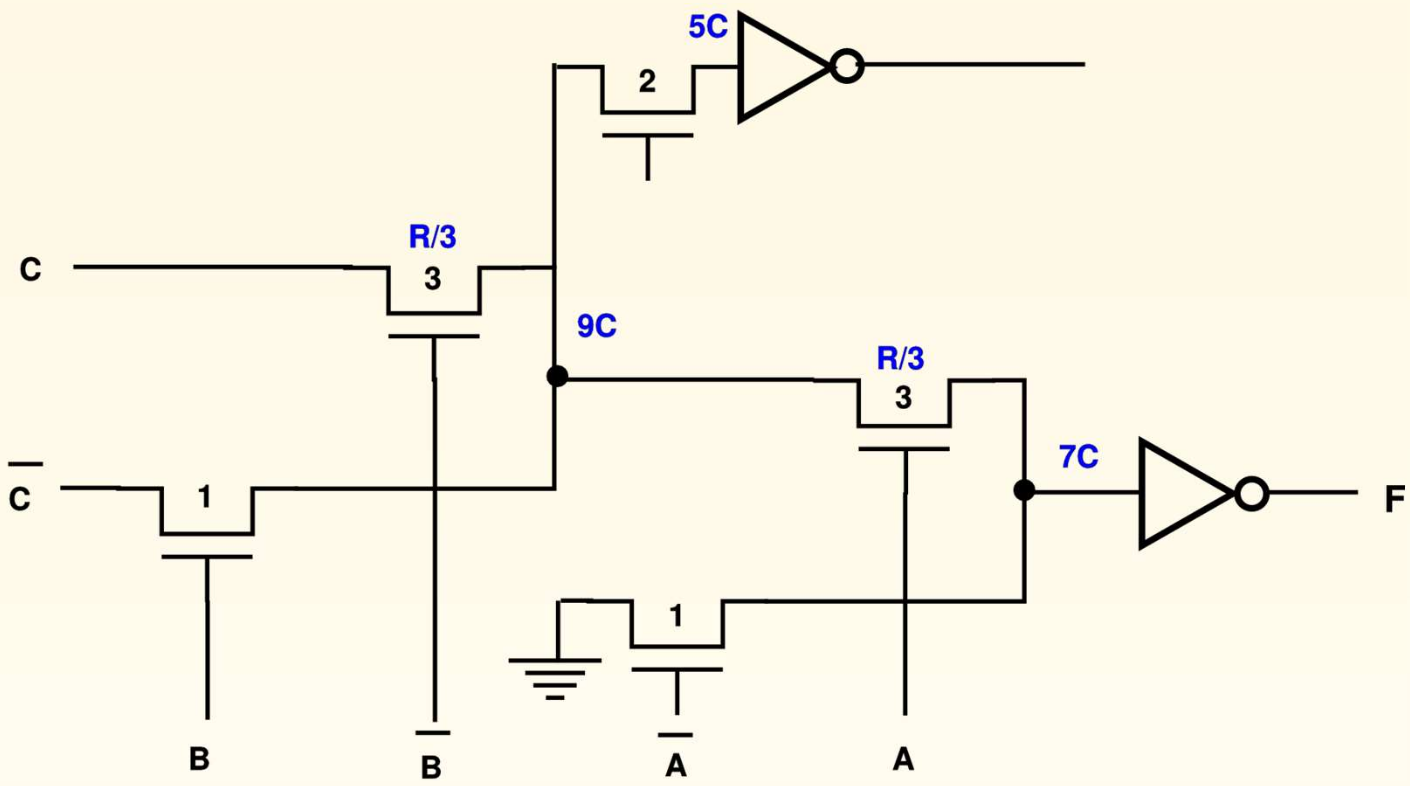
Example of Elmore Delay Calculation

Calculate the Elmore delay from C to F in the circuit. The widths of the pass transistors are shown, and the inverters have minimum-sized transistors



Example of Elmore Delay Calculation

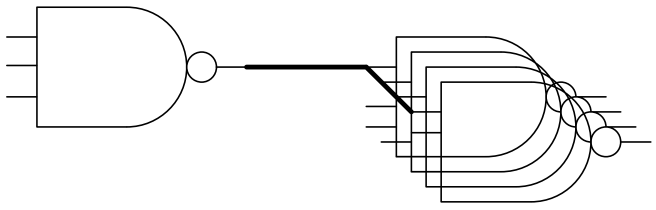
Calculate the Elmore delay from C to F in the circuit. The widths of the pass transistors are shown, and the inverters have minimum-sized transistors



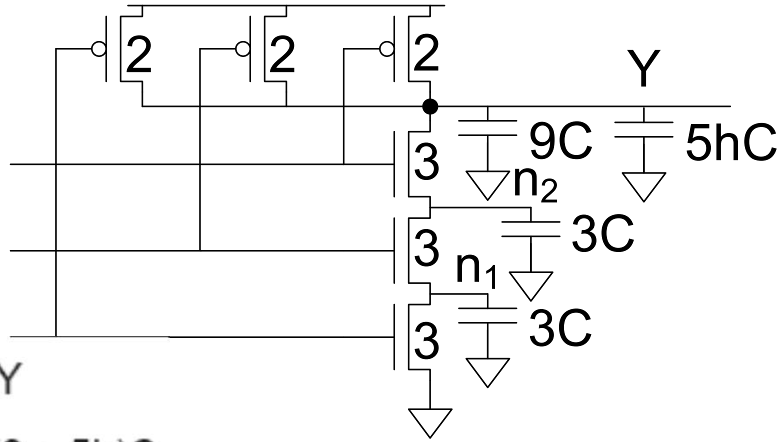
$$Delay = \frac{R}{3}9C + \frac{R}{3}5C + \left(\frac{R}{3} + \frac{R}{3}\right)7C + 3RC = 12.33RC$$

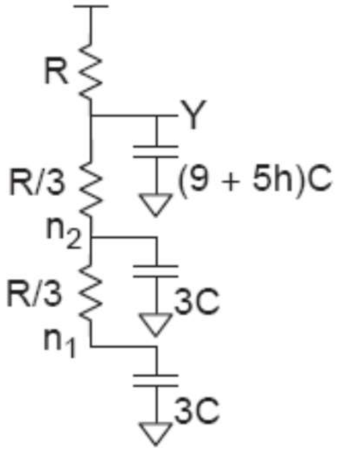
Example: 3-input NAND

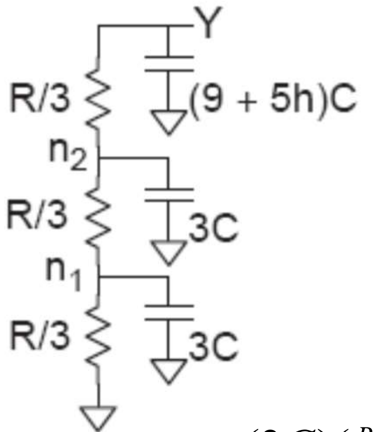
- Estimate worst-case rising and falling delay of 3-input NAND driving *h* identical gates.



h copies







$$t_{pdr} = [(9+5h)C](R) + (3C)(R) + (3C)(R) = (15+5h)RC$$

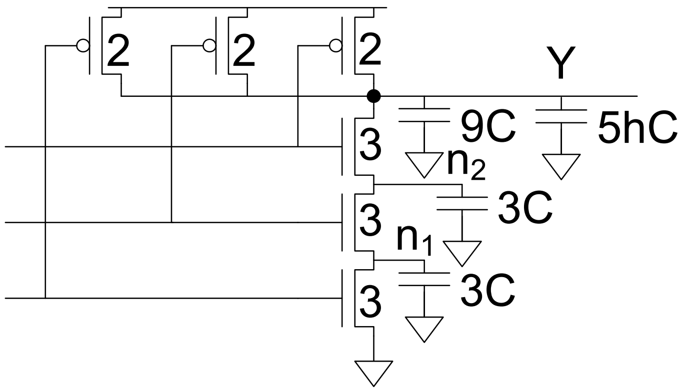
$$t_{pdf} = (3C)\left(\tfrac{R}{3}\right) + (3C)\left(\tfrac{R}{3}+\tfrac{R}{3}\right) + [(9+5h)C]\left(\tfrac{R}{3}+\tfrac{R}{3}+\tfrac{R}{3}\right) = (12+5h)RC$$

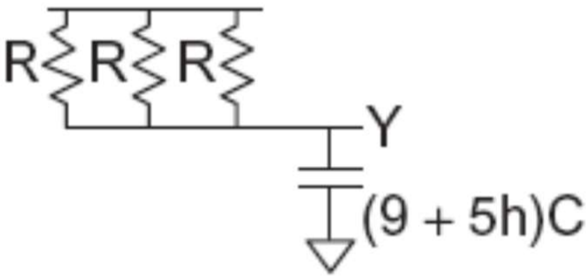
Delay Components

- Delay has two parts
  - *Parasitic delay*
    - 15 or 12 RC
    - Independent of load
  - *Effort delay*
    - 5h RC
    - Proportional to load capacitance

Contamination Delay

- Best-case (contamination) delay can be substantially less than propagation delay.
- Ex: If all three inputs fall simultaneously

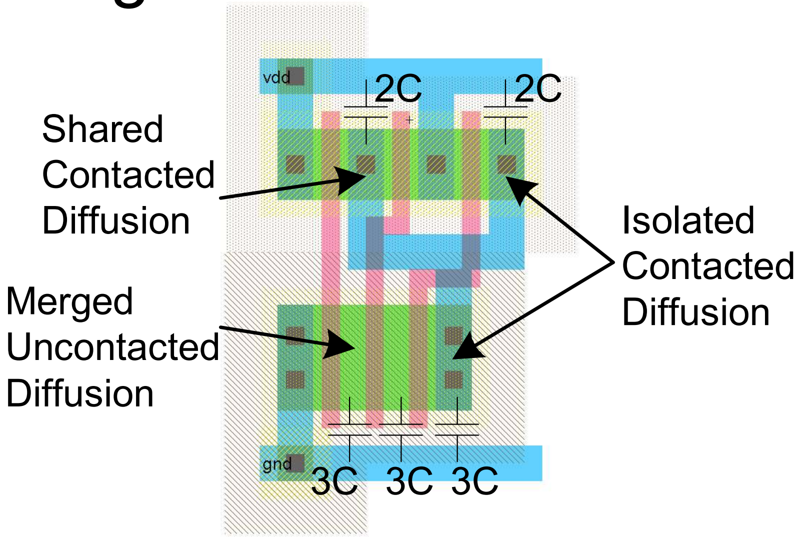


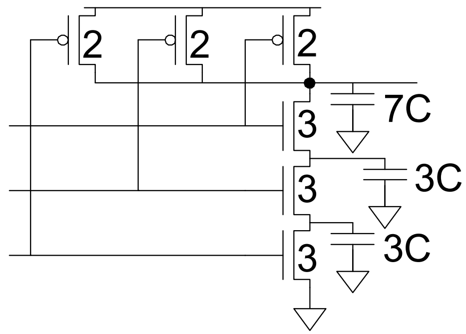


$$t_{cdr} = \left[\left(9+5h\right)C\right]\left(\frac{R}{3}\right) = \left(3+\frac{5}{3}h\right)RC$$

Diffusion Capacitance

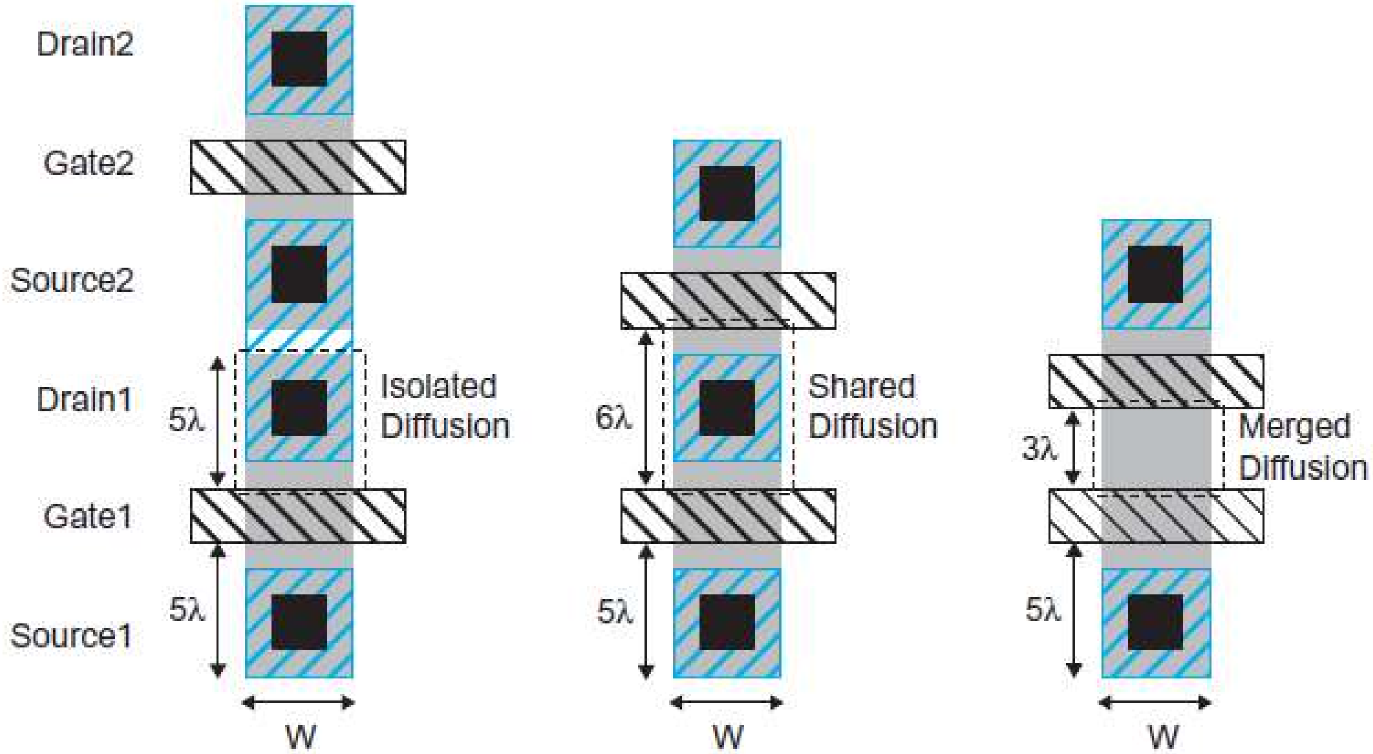
- We assumed contacted diffusion on every s / d.
- Good layout minimizes diffusion area
- Ex: NAND3 layout shares one diffusion contact
  - Reduces output capacitance by 2C
  - Merged uncontacted diffusion might help too





Isolated/Shared/Merged Diffusion

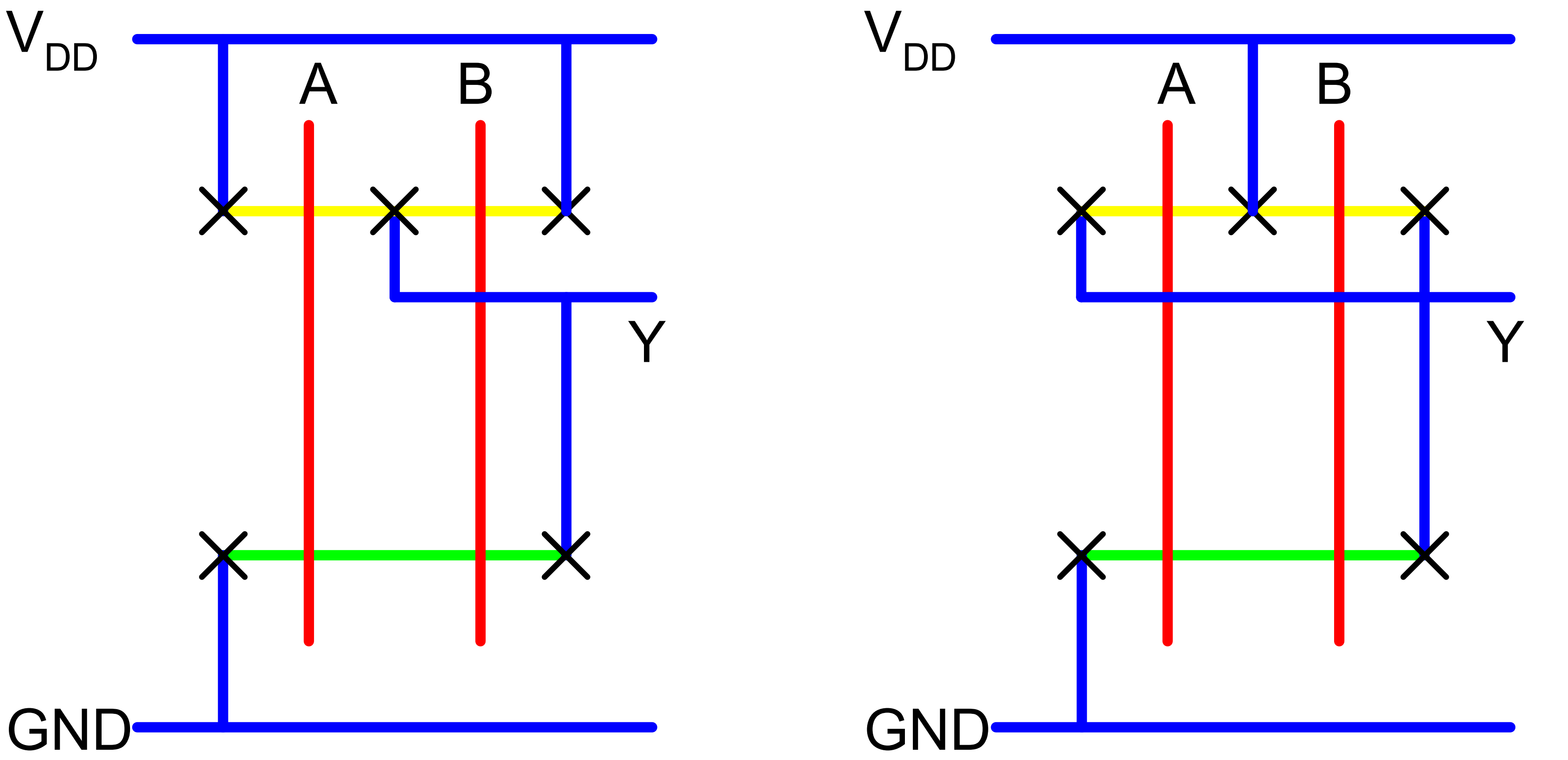
- Shared contacted diffusion can reduce the diffusion capacitance
- Un-contacted diffusion nodes can reduce more capacitance

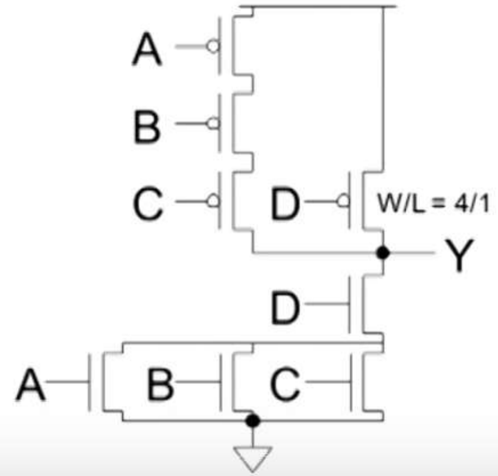


Isolated Shared Merged

Layout Comparison

- Which layout is better?



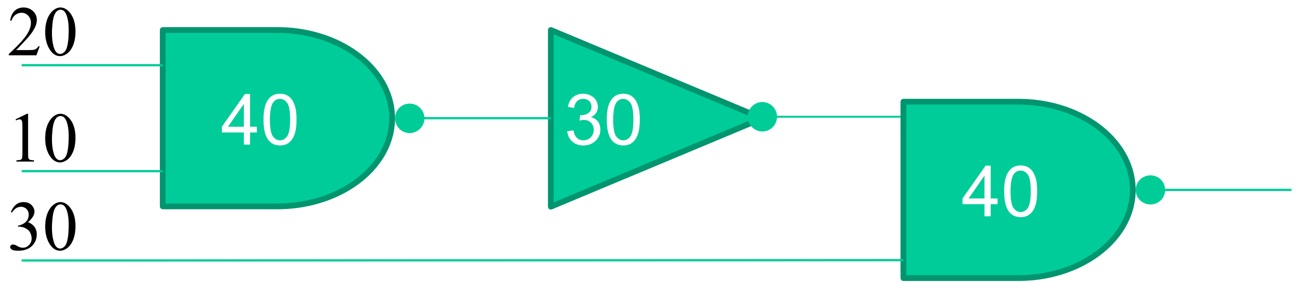


A) Size the transistors so the pull-up and pull-down resistances are matched

B) Find the input patterns that produce the longest pull-up and pull-down propagation delays

Review

1. What are t_{pdr}, t_{pdf}, t_f, t_r, t_{cdr}, t_{cdf}?
2. Calculate arrive time of the following circuit:



3. Explain the delay estimation of a fanout-of-1 inverter (slide 10)
4. Explain the t_{pdr} and t_{pdf} delay estimation of 3-input NAND driving h identical gates (slide 15).
5. Estimate delay for the gates: AOI21, OAI31
6. What is logical effort?
7. What is parasitic delay?
8. Estimate the delay of the following gate: